

Quantitative Analysis of Combined Damage to Area Targets Under Multiple Missile Strikes

Cheng-lin Zhai^{#,*} and Xiao-wei Chen[§]

[#]Beijing Institute of Petrochemical Technology, Beijing - 102 617, China

[§]Beijing Institute of Technology, Beijing - 100 081, China

*E-mail: zhaichenglin223@163.com

ABSTRACT

Quickly and accurately assessing the possible damage results of large-scale targets before being hit is the basis for formulating their protection strategies. Therefore, a quantitative method for predicting the combined damage of targets under multiple strikes is proposed in this paper. Firstly, a criterion for judging large-scale targets is constructed, and the landing points where multiple missile warheads cause the maximum damage are calculated by applying the differential evolution algorithm. Secondly, according to the maximum damage landing point and the CEP (circular error probable) of each missile warhead, the target is divided into grids, and effective grids are determined. Meanwhile, the damage radii of missile warheads are obtained by an equivalent damage calculation method. Thirdly, landing points are simulated by the Monte Carlo method, and in combination with the damage power of each missile warhead, the probability of multiple missile warheads landing at different points and the corresponding damage results can be intuitively displayed. Finally, the probability damage results of a building under different missile warhead combinations are solved, which verifies the feasibility of this method. The developed method can quickly obtain the potential damage that buildings may suffer, which can guide resource allocation and defense strategies.

Keywords: Damage assessment; Multiple missile warheads; Differential evolution algorithm; Monte Carlo method; Effective grid

NOMENCLATURE

$a_{n,m}(g)$	: The m^{th} component of the n^{th} individual in the g^{th} generation population
$A(g)$	: The g^{th} generation population
$A_n(g)$	: The n^{th} individual in the g^{th} generation population
CEP_N	: Circular error probable of the n^{th} missile warhead
f	: Evaluation function
i	: Impulse (Pa·s)
L_{xl}	: Abscissa of the left border
L_{xr}	: Abscissa of the right border
L_{yd}	: Ordinate of the lower border
L_{yu}	: Ordinate of the upper border
M	: Number of combinations of grids
N	: Number of missile warheads
NP	: Population size
P	: Ratio of percentage of damaged area
P_d	: Percentage of damaged area of buildings
P_i	: Landing probability of the i^{th} warhead
P_s	: Overpressure (Pa)
r_n	: Damage radius of the n^{th} warhead
$rand$	: Uniformly distributed random number between 0 and 1

S_i	: Damaged area caused by the i^{th} warhead
S	: Damaged area
S_d	: Lower limit of percentage of damaged area
S_{max}	: The maximum damaged area
S_u	: Upper limit of percentage of damaged area
T	: Number of all combinations
X_i	: Abscissa of the i^{th} ideal landing point
Y_i	: Ordinate of the i^{th} ideal landing point

1. INTRODUCTION

With the widespread use of missile strikes in modern warfare, how to achieve more effective missile interception and optimized resource allocation is attracting more attention. In addition, due to the limitations of large equivalent missiles, joint strikes by small equivalent missiles are often adopted when the target area is too large. Compared to the probability damage analysis of a single missile, the difficulty of probability damage analysis for multiple missiles increases significantly. To characterize the potential damage of multiple missiles to targets, the authors conduct this study with the aim of developing protection strategies and resource allocation plans.

Evaluating the damage of missiles to targets depends on missile power, landing point, landing probability, target type, damage mode, and target area. Many studies have been conducted on the relevant aspects. Certain studies concentrate on target damage. Depending on the specific focus, researchers

have evaluated target damage by comparing images captured before and after missile strikes¹. Others have examined the effectiveness of defense measures and protective materials on missile defense². While additional studies have developed risk quantification models to quantify target damage resulting from missile strikes³. The other studies focus on missile damage. These studies include the study of missile damage performance from the perspective of damage probability⁴, the development of missile system evaluation and selection algorithms from the perspective of algorithm optimization⁵, and the construction of the two-sided model for missile interception from the perspective of tactical defense⁶. Whether the subject of these studies is the target or the missile, they all focus on the damage caused by a single missile to targets, which has been the primary focus of previous research. Due to the widespread adoption of small equivalent missiles and the widespread distribution of large-scale targets, exploring the precise damage caused by a single missile strike cannot meet the needs of quantifying damage to large-scale targets. And this is very important for developing multiple missile defense strategies.

The damage studies of targets under multiple strikes are also an important component. The relevant studies mainly focus on proposing assignment algorithms with better performance based on existing algorithms for different targets or different scenarios, whose aim is to achieve the maximum damage effect⁷⁻⁹. These studies cannot provide the quantification of the damage degree and the corresponding probabilities. In terms of target damage, most studies focus on the ability to destroy targets¹⁰ and the simultaneous strike of multiple missiles¹¹. And the damaged targets are categorized into stationary¹² and maneuvering targets¹³. The targets in these studies are still small-scale targets, and the damage quantification is not applicable to large-scale targets. Additionally, few studies focus on the damage of large-scale targets, but they are limited to the calculation of ammunition consumption or the probability of destroying targets¹⁴. In summary, studies on quantifying the probability damage of large-scale targets under multiple strikes are insufficient.

In terms of quantitative analysis methods, previous studies mostly focus on the application of numerical simulation or tests to measure the damage caused by missiles¹⁵⁻¹⁶. Although the results are relatively accurate, these methods are difficult to apply to the complex and diverse combinations of damage. They are not only costly and time-consuming but also challenging to describe comprehensively and efficiently. Furthermore, using a calculation method to study the combined damage from multiple missiles to large-scale targets is mainly focused on solving the locations of multiple missiles. In the solution process, not only the damage radius of each missile needs to be given, but also the effect of landing probability is not considered¹⁷. The relevant studies only obtain the hit rate of targets and cannot address the quantification of damage degree of large-scale targets at the same time.

Advanced weapons and equipment have greatly accelerated the pace of modern warfare, which makes it more important to respond quickly and correctly, especially in missile strikes and interceptions. Many studies have been conducted on the damage to targets hit by missiles, but they

mostly focus on the damage caused by a single missile and the allocation of multiple missiles. There is a deficiency in the study of the probability of damage and the degree of damage caused by multiple missiles to targets. Current studies on the damage to targets hit by multiple missiles mostly stop at solving for the ideal landing points of multiple missiles. However, the inability to accurately describe the resulting damage to the targets makes it difficult to precisely determine appropriate resource allocation and defense measures.

Therefore, considering the deficiencies of current studies and practical needs, a study on the damage of large-scale targets hit by multiple missile warheads from the perspective of probability damage is conducted in this paper. It can quickly and intuitively quantify how much damage multiple missiles cause to large-scale targets and what the corresponding probability is. The study further explores the damage caused by multiple missiles to targets, addressing the deficiencies in multi-missile damage to some extent. Additionally, the developed methods and obtained results provide comprehensive references for resource allocation and defense strategy formulation.

2. CHARACTERIZATION OF DAMAGE POWERS

The characterization of damage powers refers to measuring the damage caused by missile warhead to targets. It mainly encompasses two aspects: the determination of large-scale targets and the quantification of equivalent damage.

2.1 Judgment Standards

Large-scale targets are those with low height and large area, which can be a single target or a combination of similar targets. Such targets typically have similar physical characteristics. In addition, the primary damage elements produced by conventional missiles that can cause area damage are blast waves and fragments. Considering the wide distribution of targets and damage tendencies of elements, buildings are selected as the target type, and blast waves are selected as the damage element type. Due to the significantly different damage patterns of low-height buildings compared to high-height buildings, they cannot be integrated for quantitative analysis. For low-height and large-area buildings, the influence of height on their damage analysis can be neglected, allowing them to be treated as two-dimensional targets. Based on this equivalence, the judgment criteria are developed.

Damage power of a missile warhead is typically measured by its damage radius. Therefore, the maximum damaged area of buildings caused by a single missile warhead is established as an indicator, and judgment criteria are developed based on the value of this indicator to determine whether buildings are likely to be subjected to multiple missile strikes. In this study, the threshold for the indicator is set at 50% of the total building area, which can be adjusted as needed.

2.2 Equivalent Damage Radii

The damage radii of missile warheads need to be calculated based on the corresponding damage powers. And the damage powers of blast waves can be measured by overpressure and impulse.

Since overpressure and impulse cannot be used directly to calculate the damage radii of missile warheads, a conversion is required. It employs the equivalent damage area for characterization. The PROBIT equation is used to describe how the damage of buildings is related to the overpressure and impulse. Under collapse level of buildings, the dependence of the percentage of damaged area on overpressure and impulse is as follows¹⁸:

$$100P_d = 50.05 - 7.31 \ln \left[\left(\frac{40000}{P_s} \right)^{7.4} + \left(\frac{460}{i} \right)^{11.3} \right] \quad (1)$$

where, P_d is the percentage of damaged area of buildings; P_s is the overpressure, Pa; i is the impulse, Pa·s.

Eqn. (1) characterizes the percentage of damaged area of buildings at any location. Now the analyzed object is converted into missile warheads, and damage radii are solved by the equivalent calculation method¹⁸. The basic idea is as follows:

- Divide the building area into grids according to the accuracy requirements
- Calculate the percentage of damaged area at each grid in turn
- Solve the total damaged area of buildings based on the percentage of damaged area at each grid
- Calculate the equivalent damage radii of missile warheads based on geometric relationships.

Using this method to calculate damage radii can greatly reduce the error caused by different percentages of damaged area. By utilizing the damage radius of each missile, subsequent calculations of the combined damage area can be carried out.

3. CALCULATION OF DAMAGED AREA

After clarifying damage powers, the calculation of the damaged area follows. Since buildings are simplified into polygons and the damage scope of a missile warhead is characterized by a circle, the calculation of the damaged area is the classified quantification of the overlapping area of the two.

3.1 Overlapping area between a single circle and a polygon

According to the positional relationship between a circle and a polygon, it can be classified into two cases: the center of the circle is either inside or outside a polygon, as shown in Fig.1. The calculation of overlapping area between a circle and a polygon can be transformed into the calculation of overlapping area between a circle and directed triangles. Based on Stewart's theorem¹⁹, the overlapping area under each positional relationship can be solved separately. Since the direction of directed triangles needs to be specified, the clockwise direction is specified as positive. Based on the direction specification, it can be determined that all of triangles in Fig.1(a) are positive triangles, and the corresponding overlapping areas are positive. In Fig. 1(b), $\triangle AOB$, $\triangle BOC$, $\triangle COD$, $\triangle DOE$ and $\triangle EOF$ are positive triangles, and the corresponding overlapping areas are positive. $\triangle FOA$ is a negative triangle, and the corresponding overlapping area is negative.

A polygon is split into directed triangles, and the overlapping area between each directed triangle and the circle

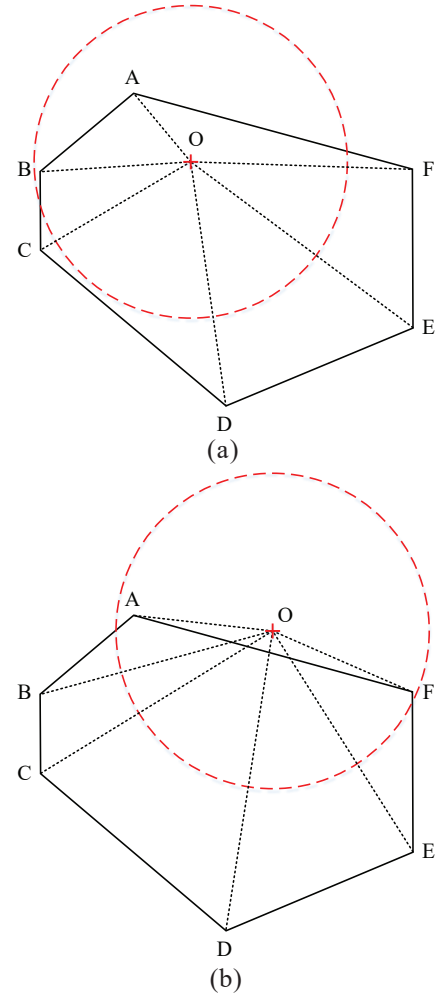


Figure 1. Positional relationship between a circle and a polygon; (a) Center of circle is inside a polygon; and (b) Center of circle is outside a polygon.

is calculated in turn. Then the positive and negative overlapping areas are accumulated to obtain the total overlapping area.

3.2 Overlapping Area Between Multiple Circles and a Polygon

Different from the calculation of overlapping area between a circle and a polygon, it is necessary to analyze the positional relationship among multiple circles when calculating the overlapping area between multiple circles and a polygon. The positional relationship of multiple circles in a plane can be classified into two cases, which are no intersection among circles and intersection among circles. For the case of no intersection among circles, it is required to solve the overlapping area between each circle and the polygon first. Then, each overlapping area is accumulated to obtain the overlapping area between multiple circles and a polygon.

For the case of intersection among circles, two circles are selected for analysis, as shown in Fig.2. First, the polygon is divided into two parts by the line where intersections are located. Then it is required to solve the overlapping area between each circle and the corresponding part. Finally, the two overlapping areas are summed to obtain the overlapping area between two intersecting circles and a polygon. For

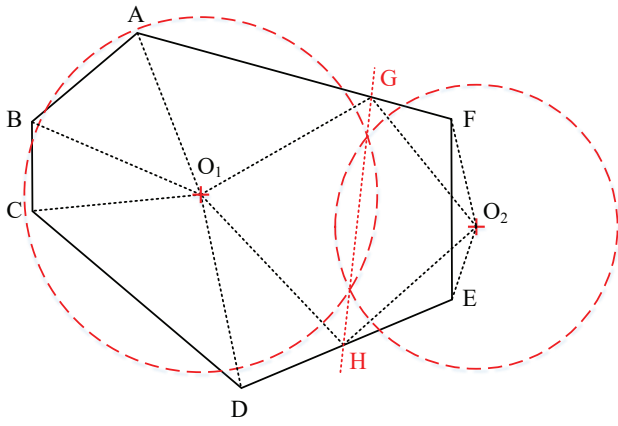


Figure 2. A positional relationship between two intersecting circles and a polygon.

three or more intersecting circles, the calculation method of overlapping area is similar to that of two intersecting circles.

4. PROGRAMMING OF IDEAL LANDING POINTS

Multiple strikes usually take the locations where multiple missile warheads can cause the largest damage to the buildings as the aiming points. And those aiming points are named ideal landing points in this paper.

The programming of ideal landing points is a nonlinear and multi-peak function optimization problem, and it is very difficult to obtain the global optimum. Current accurate calculation methods, such as the dynamic programming algorithm and branch and bound algorithm²⁰⁻²¹, are very complicated, and cannot meet the requirement of quickly obtaining ideal landing points. Moreover, individual-based heuristic algorithms, such as the simulated annealing algorithm and tabu search algorithm²², can easily obtain locally optimal solutions. Therefore, on the basis of comprehensively comparing the advantages and disadvantages of existing methods, the differential evolution algorithm²³ is selected as the method for calculating ideal landing points due to its concise structure, parallel efficiency, fast convergence and reliability.

4.1 Coding

For the case of using N missile warheads, the coordinates of the ideal landing points are coded, and the individual code with a length of $2N$ is obtained as:

$$(X_1, Y_1, X_2, Y_2, \dots, X_i, Y_i, \dots, X_N, Y_N) \quad (2)$$

where, X_i is the abscissa of the i^{th} ideal landing point; Y_i is the ordinate of the i^{th} ideal landing point; N is the number of missile warheads.

4.2 Population Size and Initialization

The population size is the number of individuals. An appropriate population size not only ensures the search ability of the algorithm, but also reduces the convergence time of the algorithm. The interval of population size is usually [50,200], and the population size selected in this paper is 100.

$$\begin{cases} A(g) = (A_1(g), A_2(g), \dots, A_n(g), \dots, A_{NP}(g)) \\ A_n(g) = (a_{n,1}(g), a_{n,2}(g), \dots, a_{n,m}(g), a_{n,2N}(g)) \end{cases} \quad (3)$$

where, NP is the population size; $A(g)$ is the g^{th} generation population; $A_n(g)$ is the n^{th} individual in the g^{th} generation population; $a_{n,m}(g)$ is the m^{th} component of the n^{th} individual in the g^{th} generation population.

The range of individual components should be delimited first. And the meaning of individual components is the coordinates of ideal landing points. Draw a rectangle that completely covers the polygonal buildings. The abscissae of the left and right borders of the rectangle are respectively marked as L_{xl} and L_{xr} , which satisfy $L_{xl} < L_{xr}$. The ordinates of the upper and lower borders of the rectangle are respectively marked as L_{yu} and L_{yd} , which satisfy $L_{yd} < L_{yu}$.

Then the code of each individual in the initial population is randomly and uniformly generated within the rectangle. Since the coordinates of each ideal landing point form a two-dimensional vector, the code of the n^{th} individual $A_n(0)$ is:

$$\begin{cases} x_n = L_{xl} + (L_{xr} - L_{xl}) \times rand \\ y_n = L_{yd} + (L_{yu} - L_{yd}) \times rand \end{cases} \quad (4)$$

where, $rand$ is a uniformly distributed random number between 0 and 1.

The code of the n^{th} individual $A_n(0)$ in the initial population can be obtained by Eqn.(4), and then the initial population is generated according to population size.

4.3 Evaluation Function

Based on the characteristics of programming ideal landing points, the damaged area of buildings is selected as the index, which is the basis for constructing an evaluation function.

The purpose of programming the ideal landing points of multiple missile warheads is to obtain the maximum damaged area of polygonal buildings. When N missile warheads are used, the corresponding damage radii are $r_1, r_2, \dots, r_n$, respectively. And the corresponding coordinates of the N ideal landing points are $(X_1, Y_1), (X_2, Y_2), \dots, (X_N, Y_N)$, respectively. Then the maximum damaged area of each individual is calculated by the method in Section 3, and the expression is as follows.

$$S_{max} = \sum_{i=1}^N S_i(X_i, Y_i), (i=1, 2, \dots, N) \quad (5)$$

where, S_{max} is the maximum damaged area caused by N missile warheads; $S_i(X_i, Y_i)$ is the damaged area caused by the i^{th} missile warhead under the premise of the maximum damaged area.

Based on Eqn. (5), the evaluation function for solving the ideal landing points is:

$$f = S = \sum_{i=1}^N s_i(x_i, y_i), (i=1, 2, \dots, N) \quad (6)$$

where, f is the evaluation function; S is the damaged area caused by N missile warheads; s_i is the damaged area caused by the i^{th} missile warhead; x_i and y_i are the abscissa and ordinate of landing point of the i^{th} missile warhead, respectively.

5. CALCULATION OF COMBINED PROBABILITY DAMAGE

Missile warheads will randomly land on the plane containing buildings due to interference factors. But the landing points of each warhead will follow a distribution pattern centered around the ideal landing point. Due to the lack

of detailed conditions, the two-dimensional normal distribution model²⁴, which is the most common distribution, is adopted in this paper. The standard deviation can be directly related to CEP²⁵, which is the basis of calculating the probability of landing points.

When solving the combined damage of multiple missile warheads, it is necessary to design a method to comprehensively characterize the landing probability and damage degree.

5.1 Principle of Designed Method

An area must be specified before solving for the probability of landing points. Furthermore, as the probability solutions for each warhead are independent of each other, then the plane containing the building is divided into grids. A grid is regarded as the basic unit for solving the probability of the landing points. Monte Carlo method²⁶ is used to simulate landing points, and landing point probability of each warhead is solved by counting the number of landing points in grids. Based on the landing point probability of each warhead, the combined landing probability is calculated. In addition, the center point of each grid is regarded as the calculation point where the damaged area is calculated. Part of a polygonal building, and the corresponding diagram are shown in Fig. 3.

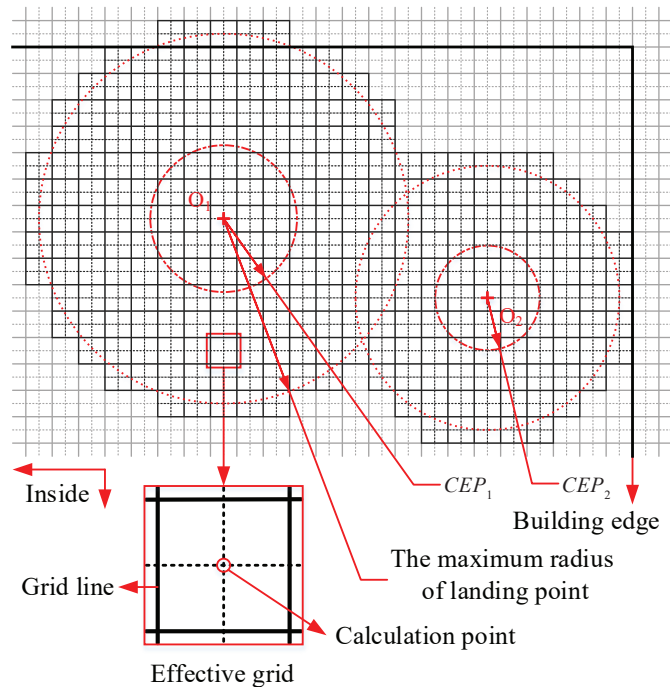


Figure 3. Diagram of the probability of landing point calculation.

Each small square formed by four solid lines is a grid, which is used as a basic unit for calculating the probability of landing points. Each intersection formed by the dashed lines is a calculation point. The black grids centered on ideal landing points are effective grids, which are determined by the maximum landing radius.

5.2 Process of Solution

The CEPs and damage radii of N missile warheads are $[CEP_1, CEP_2, \dots, CEP_N]$ and $[r_1, r_2, \dots, r_N]$, respectively. Based

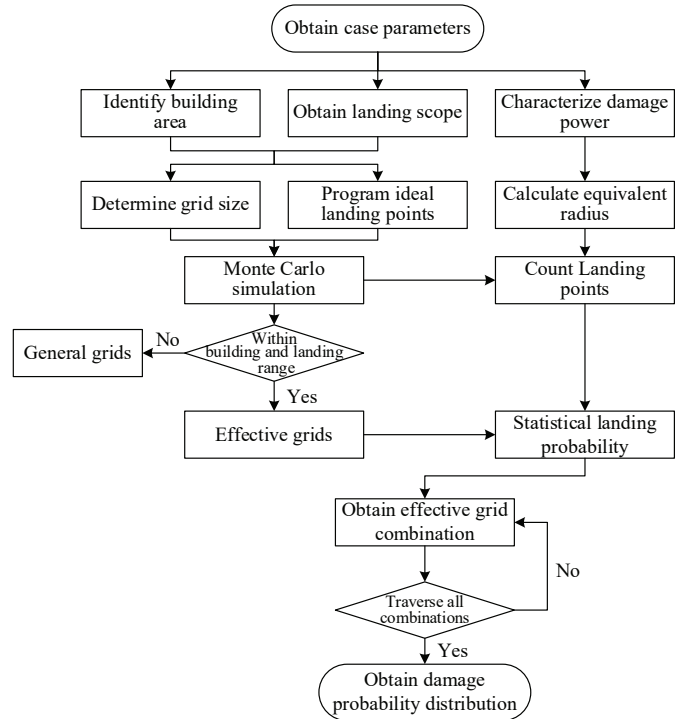


Figure 4. Calculation process for the combined damage probability.

on the principle of the designed method, the calculation process for the combined damage probability is shown in Fig. 4.

Different landing probabilities correspond to varying damage probabilities. Based on the damage radii of N warheads, the damaged areas of buildings corresponding to different combinations of landing point can be calculated. By integrating the damage probabilities and damaged areas for all landing point combinations, the probability damage results for the buildings can be obtained.

Statistical analysis is carried out on the combined landing probability and the corresponding damaged area in all combinations. The combined landing probability is the product of the landing probability of each warhead. Since the area of different buildings is different, it is characterized by the percentage of damaged area. Based on this rule, the expression is given.

$$P[S_d, S_u] = \frac{\sum_{j=1}^M \prod_{i=1}^N P_i[S_d, S_u]}{\sum_{j=1}^T \prod_{i=1}^N P_i} \quad (7)$$

where, S_d and S_u are the lower limit and upper limit of percentage of damaged area, respectively; $P[S_d, S_u]$ is the ratio of percentage of damaged area satisfying the given range of $[S_d, S_u]$; $P_i[S_d, S_u]$ is the landing probability of the i^{th} warhead satisfying the given range of $[S_d, S_u]$; M is the number of combinations of grids satisfying the given range of $[S_d, S_u]$; P_i is the landing probability of the i^{th} warhead; T is the number of all combinations.

6. CASE STUDY

6.1 Parameters

The outer contour of a building extracted from the satellite image is shown in Fig. 5, and the parameters of missile warheads are shown in Table 1.

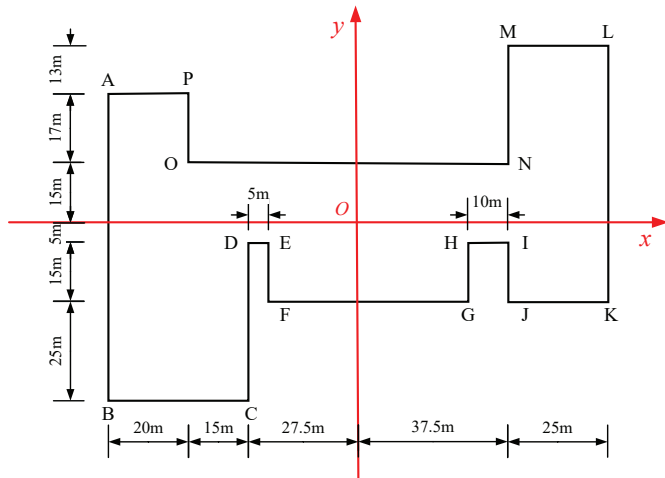


Figure 5. Outer contour of a building.

Table 1. Parameters of missile warheads

Type	Mass (kg)	Explosive heat (kJ/kg)	CEP (m)
A	150	5673	10
B	100	5673	5

6.2 Ideal Landing Points

Through the equivalent calculation method of damaged area, the percentage of damaged area of the building caused by the two types of warhead is calculated separately, and the results are shown in Fig.6.

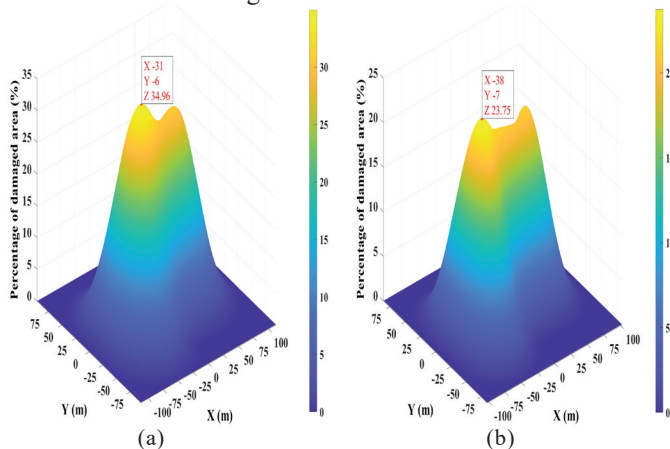


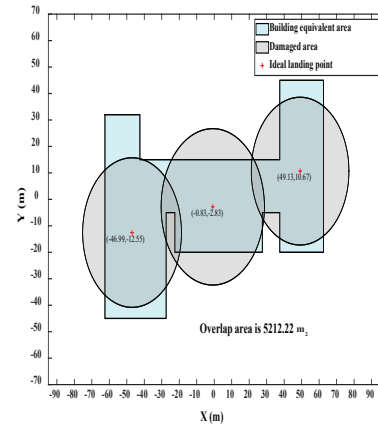
Figure 6. Damage maps of the building under the two types of warheads strike.

Figure 6 shows that the maximum damaged area caused by the two types of warheads is 34.96 % and 24.75 %, respectively. According to the judgment standards in Section 2.1, it can be determined that this building is a large-scale building.

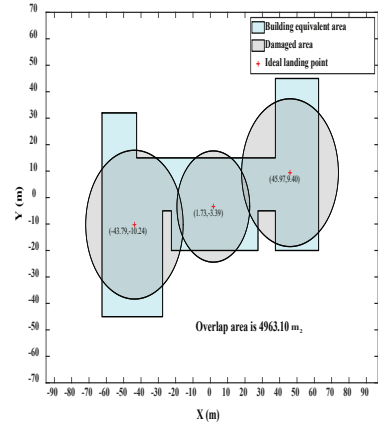
Since the partial combined damage caused by two warheads still does not meet the judgment standards, the damage caused by three warheads is discussed in this paper. And there are four combinations of warheads, as shown in Table 2.

Table 2. Combinations of warheads

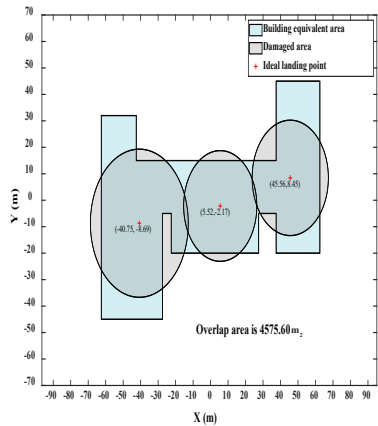
Number	Combination		
Case 1	A	A	A
Case 2	A	A	B
Case 3	A	B	B
Case 4	B	B	B



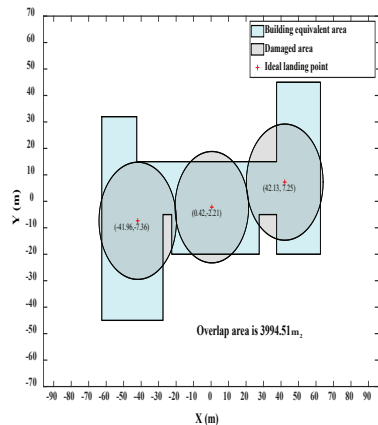
Case 1



Case 2



Case 3



Case 4

Figure 7. Ideal landing points for four cases.

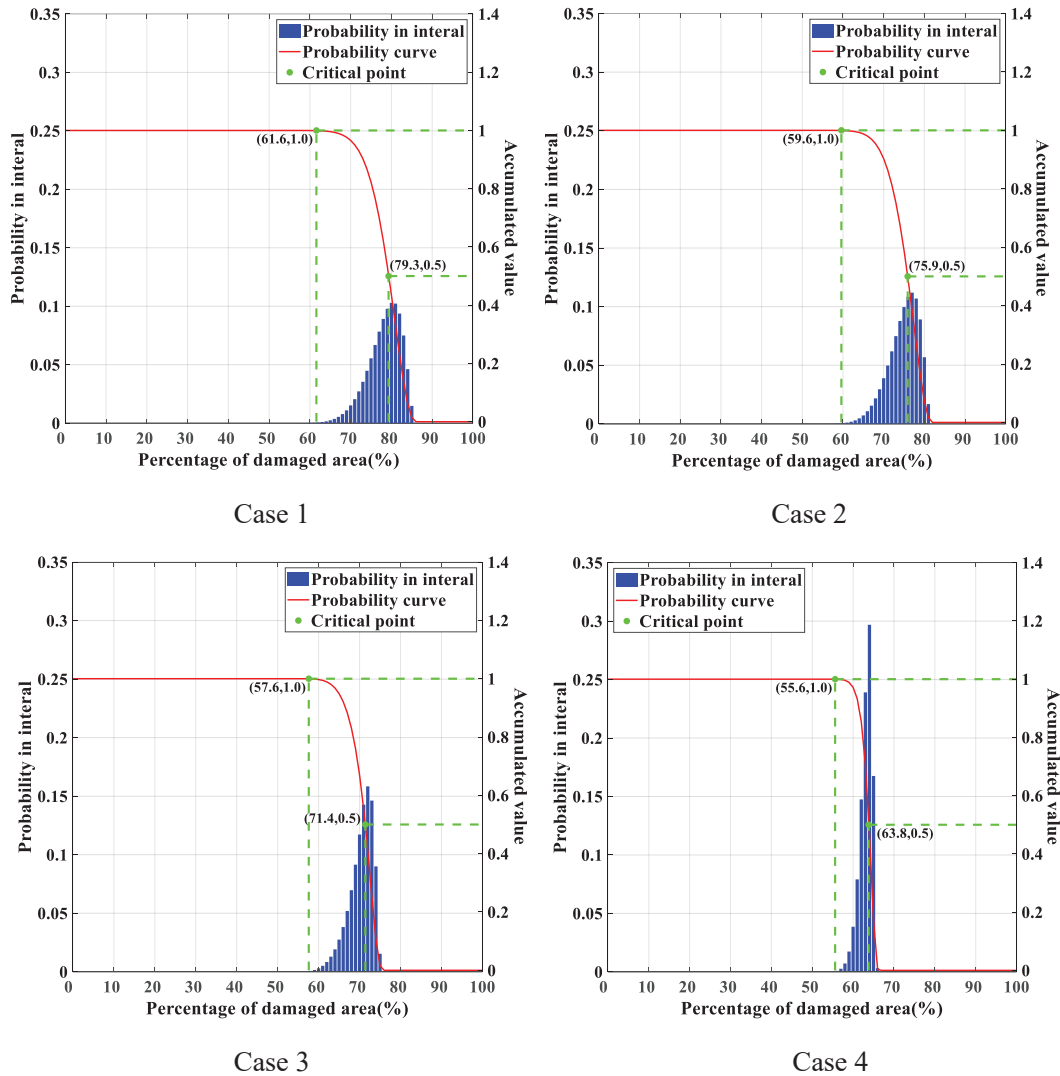


Figure 8. Probability distributions and probability curves under four cases.

Based on the percentage of damaged area for the two types of warheads at each calculation point, the corresponding damage radii are calculated. Then, the method in Section 4 is used to calculate the locations of ideal landing points for the four cases in sequence. The results are shown in Fig. 7.

The maximum damaged areas of Case 1 to 4 are 5212.22 m², 4963.10 m², 4575.60 m² and 3994.51 m², respectively. By dividing the total area of 6115 m², the maximum percentages of damaged area under Case 1 to 4 are 85.24 %, 81.16 %, 74.83 % and 65.32 %, respectively.

6.3 Probability Damage of Buildings

According to the CEPs in Table 1, the standard deviations corresponding to the Type A and Type B warheads are calculated, which is 8.50 m and 4.25 m, respectively. Then the calculation method of probability damage in Section 5 is employed, and under the premise of taking 1 % as the scale, statistical calculations are performed on Cases 1 to 4. Finally, the probability distributions and probability curves of the percentage of damaged area are obtained, as shown in Fig.8. The blue histograms in Fig. 8 represent the damage probability, and its longitudinal axis is on the left. The red curves in

Fig. 8 represent the cumulative probability greater than the corresponding scale, and its longitudinal axis is on the right.

Figure 8 shows that the damage probability ranges of Cases 1 to 4 are 61.6 %-85.2 %, 59.6 %-81.2 %, 57.6 %-74.8 % and 55.6 %-65.32 %, respectively. After analysis, the following conclusions can be drawn:

- The starting points of damage probability for Cases 1 to 4 are 61.6 %, 59.6 %, 57.6 %, and 55.6 %, respectively. And these values are the minimum of percentage of damaged area under the four cases
- Since the CEP of the Type B warhead is greater than the CEP of the Type A warhead, the damage probability histogram of Case 1 has the widest distribution range, and the damage probability at 1 % interval is relatively small. The distribution range of the damage probability histogram of Case 4 is the most concentrated, and the damage probability at 1 % interval is relatively large.

Different combinations correspond to different probability damage results, and different probability damage results correspond to different disposal strategies. In order to display the differences among the combinations, the probability curves of the four cases are plotted together, as shown in Fig. 9.

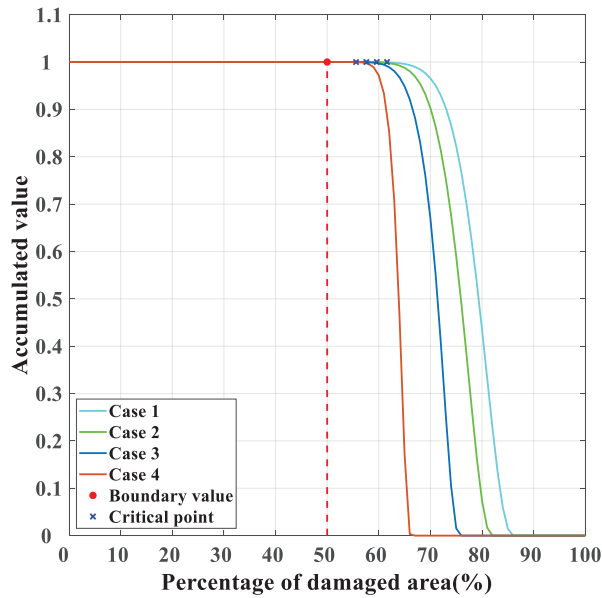


Figure 9. Combination diagram of probability curves for four cases.

Previous studies have rarely focused on quantifying the damage of multiple missiles to large-scale targets, and similar studies have not been conducted on quantifying the probability of damage of multiple missiles. Therefore, there are no relevant results available for comparison.

7. CONCLUSION

A quantitative analysis of the damage of area targets hit by multiple missiles is performed in this paper, and the following conclusions are obtained.

Through the probability damage results combined by landing probability and damaged area, any damage probability and percentage range of damaged area can be quickly obtained. It directly provides guidance for formulating relevant strategies.

Although three missile warheads of two types are selected for probability damage calculation, the type, number and landing point distribution can be adjusted arbitrarily.

Because the percentage of damaged area of buildings and landing probability of missile warheads are solved based on grids, the object that can be calculated by this method also includes an area containing multiple buildings.

The application scope of method is limited to low-height and large-area building targets, and the effects of height are neglected. It is necessary to improve the damage relationship in the calculation method, which can be studied in the future.

REFERENCES

- Xu Y, Yu Q, Wang Y, Xiao J, Zhou Z, Lu H. Ground target detection and damage assessment by patrol missiles based on YOLO-VGGNet. *Appl Sci*. 2022;12(19):9484. doi:10.3390/app12199484
- Häring I, Pfeiffer M, Vogelbacher G, Stottmeister A, Restayn EM, Ross K, Ramin MV. Risk and resilience analysis of public civil buildings against shelling with explosive sources in urban contexts. *Eur J Secur Res*. 2020;5(2):311-347. doi:10.1007/s41125-019-00046-9
- Zhai C, Chen X. Damage assessment of the target area of the island/reef under the attack of missile warhead. *Def Technol*. 2020;16(1):18-28. doi:10.1016/j.dt.2019.06.022
- Dantas JP, Costa AN, Geraldo D, Maximo MR, Yoneyama T. PoKER: a probability of kill estimation rate model for air-to-air missiles using machine learning on stochastic targets. *J Def Model Simul Appl*. 2025;17(1):1-13. doi:10.1177/15485129241309675
- Jin T, He S, Li H, Zheng X. Evaluation model and exact optimization algorithm in missile-target assignment. *J Guid Control Dyn*. 2023;46(9):1834-1841. doi:10.2514/1.G007354
- Wade BM. A multi-objective optimization of ballistic and cruise missile fire plans based on damage calculations from missile impacts on an airfield defended by an air defense artillery network. *J Def Model Simul Appl*. 2019;16(2):103-117. doi:10.1177/1548512918788503
- Li J, Wu G, Wang L. A comprehensive survey of weapon target assignment problem: Model, algorithm, and application. *Eng Appl Artif Intell*. 2024;137:109212. doi:10.1016/j.engappai.2024.109212
- Dirik N, Hall SN, Moore JT. Maximizing strike aircraft planning efficiency for a given class of ground targets. *Optim Lett*. 2015;9(8):1729-1748. doi:10.1007/s11590-014-0844-5
- Hocaoğlu MF. Agent-based target evaluation and fire doctrine: an aspect-oriented programming view. *J Def Model Simul Appl*. 2022;19(1):107-121. doi:10.1177/15485129211040369
- Gao K, Xiao H, Qu L, Wang S. Optimal interception strategy of air defence missile system considering multiple targets and phases. *Proc Inst Mech Eng Part O J Risk Reliab*. 2022;236(1):138-147. doi:10.1177/1748006X211022111
- Yan P, Fan Y, Liu R, Wang M. Distributed target-encirclement guidance law for cooperative attack of multiple missiles. *Int J Adv Robot Syst*. 2020;17(3):172988142092914. doi:10.1177/1729881420929140
- Zhou J, Yang J. Distributed guidance law design for cooperative simultaneous attacks with multiple missiles. *J Guid Control Dyn*. 2016;39(10):2439-2447. doi:10.2514/1.G001609
- Zhou J, Lü Y, Li Z, Yang J. Cooperative guidance law design for simultaneous attack with multiple missiles against a maneuvering target. *J Syst Sci Complex*. 2018;31(1):287-301. doi:10.1007/s11424-018-6317-7
- Li Q, Chen G, Li Z, Xu L, Wei Z. Research on missile consumption in wartime with the airport runway as the target. *Rev Int Metodos Numer Calc Diseno Ing*. 2021;37(3):1-15. doi:10.23967/j.rimni.2021.09.007
- Wade BM. Creating surrogate models for an air and missile defense simulation using design of experiments and neural networks. *J Def Model Simul Appl*. 2021;18(4):273-284. doi:10.1177/1548512919877987
- Nishida A, Kang Z, Nagai M, Tsubota H, Li Y. Evaluation of local damage to reinforced concrete panels subjected

- to oblique impact by soft missile. *Nucl Eng Des.* 2019;350:116-127.
doi:10.1016/j.nucengdes.2019.04.024
17. Zhao X, Zhao X, Shen Q, Yang J, Yang W, Xie X. Application of particle swarm optimization algorithm in area target fire assignment. In: *Proc 9th Int Conf Modelling, Identification and Control (ICMIC)*; 2017 Jul 10–12; Kunming, China. p. 770-775.
doi:10.1109/ICMIC.2017.8321558
 18. Zhai C, Chen X. Probability damage calculation of building targets under the missile warhead strike. *Reliab Eng Syst Saf.* 2020;202:107030.
doi:10.1016/j.ress.2020.107030
 19. Gelişgen Ö, Kaya R. The alpha-version of the Stewart's theorem. *Demonstr Math.* 2013;46(4):795-808.
doi:10.1515/dema-2013-0481
 20. Satic U, Jacko P, Kirkbride C. A simulation-based approximate dynamic programming approach to dynamic and stochastic resource-constrained multi-project scheduling problem. *Eur J Oper Res.* 2024;315(2):454-469.
doi:10.1016/j.ejor.2023.10.046
 21. Song X, Li Z, Chen P, Zhang K, Zou Y, He W. Modeling and intelligent optimization of the deployment system of defensive weapons based on the interior point method with barrier functions. *Chin J Eng.* 2024;46(1):157-165.
doi:10.13374/j.issn2095-9389.2022.12.26.002
 22. Rossi A, Singh A. Minimum-cardinality global defensive alliances in general graphs. *Ann Oper Res.* 2025;349:1891-1931.
doi:10.1007/s10479-025-06571-2
 23. Chai X, Zheng Z, Xiao J, Yan L, Qu B, Wen P, Wang H, Zhou Y, Sun H. Multi-strategy fusion differential evolution algorithm for UAV path planning in complex environment. *Aerosp Sci Technol.* 2022;121:107287.
doi:10.1016/j.ast.2021.107287
 24. Zhai C, Chen X. Simulation model for studying the effect of function distribution on the evaluation of building damage caused by missile attack. *Def Sci J.* 2023;73(5).
doi:10.14429/dsj.73.18317
 25. Ma J. Constrained target clustering for military targeting process. *Def Sci J.* 2017;67(5):523-528.
doi:10.14429/dsj.67.10441
 26. Taneja SB, Jaiswal S, Bisht S, Jindal V, Bedi P. MOP-N: a hybrid AI model for automated deployment of guns in dynamic wargaming scenario. *Def Sci J.* 2025;75(2).
doi:10.14429/dsj.20246

ACKNOWLEDGEMENT

This work was supported by the National Natural Science Foundation of China (11627901, 11872118).

CONTRIBUTORS

Dr Chenglin Zhai obtain his PhD from Beijing Institute of Technology and working as an Assistant Professor at Beijing Institute of Petrochemical Technology, China. His research interests include: Damage assessment, risk quantification, emergency management.

In the current study, he carried out approach design, overall calculation and preparation of the manuscript.

Prof Xiaowei Chen obtained his PhD from Nanyang Technological University. He is currently a Professor and a doctoral supervisor at Beijing Institute of Technology, China. His research interests include: Hypervelocity impact, armour-piercing and penetration, damage assessment.

In the current study, he provided guidance and reviewed the manuscript.